Nguyen Phan Duy Dang

Software Engineer

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No. 2251, Huynh Tan Phat Street, Nha Be

CAREER OBJECTIVE

A performance-driven Software Engineer leveraging a robust foundation in Object-Oriented Programming (OOP), high-throughput data processing, and concurrent programming to build scalable, clean enterprise services. Proficient in designing layered architectures, optimizing memory footprints, and implementing asynchronous workflows. Aiming to master backend technologies and contribute to high-performance system engineering within the **Java/Spring Boot** ecosystem.

EDUCATION

Ton Duc Thang University 10/2018 - 2/2024

Control Engineering and Automation

Current achievement : Good (GPA 7.8)

FUNiX - Online 2025 - 2025

Automotive Embedded Systems Fundamentals

FUNiX Online– Certificates: Web Development Foundations (HTML, CSS, JavaScript)

PROFESSIONAL EXPERIENCE

TID Company Ltd. Apr 2022 - Aug 2022

Software Engineer

Developed automation scripts for CAD workflows and programmed configurations for electrical control systems.

Northstar Adjuster JSC Nov 2023 - May 2024

Software Engineer

Built data processing tools to automate the analysis and generation of disaster compensation claims.

KTT Investment Group JSC Mar 2024 - Sep 2024

Software Engineer

Developed digital tools/models to automate MEP quantity take-offs and track construction project progress.

Industrial Environmental Solutions Co., Ltd. Nov 2024 - Mar 2025

Software Engineer

Programmed 3D piping model optimizations and managed digital databases for final project documentation

FPT Software Oct 2025 - Now

Software Engineer

Technologies: Java Core (Java 17), Java Collection Framework (ConcurrentHashMap, ArrayDeque), Spring Boot, Event-Driven Architecture, JSON, Multithreading (ExecutorService, CompletableFuture).

Description: Developed a high-performance orchestration engine designed for complex data streaming, high-throughput payload parsing, and asynchronous metadata mediation, simulating enterprise-grade application workloads.

Key Responsibilities & Achievements:

- **Core Architecture:** Designed and implemented a robust, decoupled Layered Architecture using MVC patterns, ensuring strict separation of concerns between core backend business logic (Services) and data consumption layers.
- **Asynchronous & Event-Driven Processing:** Engineered asynchronous, event-driven signaling mechanisms using **Java CompletableFuture** and **ExecutorService** to handle real-time data synchronization and concurrent system states without blocking execution threads.
- **Data Parsing & Validation:** Built enterprise-grade structured data verification workflows utilizing **Java Functional Programming (Stream API & Lambdas)**, integrating strict input validation to safely process high-throughput raw data streams.
- **Memory & Resource Optimization:** Optimized data buffering strategies under heavy processing loads by leveraging optimized Java Collections (ConcurrentHashMap, LinkedBlockingQueue) to enforce a predictable JVM Heap footprint.
- **Performance Tuning & Code Quality:** Spearheaded code refactoring initiatives by integrating **SonarLint/SonarQube** analysis into the development workflow; optimized complex SQL join queries and Hibernate fetching strategies (LAZY vs EAGER loading), reducing database response latency by 25% and resolving potential memory leaks in automated microservices.
- **Persistence & Cache Management:** Developed a localized persistence and abstract caching layer utilizing JSON structures to accelerate runtime state reload operations and reduce database overhead.

ACTIVITIES

Study Project 10/2018 - 2/2024

Personal Research

Automated IoT Smart Data System | *Java/JavaScript, Node-RED, SQL/Google Sheets, ESP32, C++.*

- **Description**: Built an automated IoT tracking system that streams real-time environmental data to external endpoints.
- **Impact**: Engineered the data pipeline with asynchronous messaging protocols, parsing structured data payloads before persisting them into relational models.

Smart Vehicle Telemetry & Notification Pipeline | ***Java 17, Spring Boot, Spring Kafka, Redis, PostgreSQL, Docker*** 7/2025 - 9/2025

Personal Research

***Description*:** Developed a high-throughput backend data pipeline designed to ingest, process, and cache real-time telemetry streaming data (speed, engine temperature, fuel metrics) from vehicle simulator clusters.

***Key Features*:**

- **Message Broker Integration:** Utilized **Spring Kafka** to build scalable consumers that efficiently handle high-volume event streams asynchronously, preventing data loss during peak traffic.
- **Distributed Caching:** Implemented a multi-tier caching strategy using **Redis** to store active vehicle states, cutting down redundant primary database read queries by over 40%.
- **Data Persistence:** Designed a relational schema in **PostgreSQL**, leveraging Spring Data JPA for batch inserts and structured indexing to handle historical logs.
- **Containerization:** Containerized the entire microservice ecosystem (Application, Kafka, Redis, DB) using **Docker** and **Docker Compose**, ensuring seamless local environment replication and scalable cloud readiness.

SKILLS

• **Programming Languages:**	Java (Core, Functional Programming/Stream API, Multithreading/Concurrency), C/C++, SQL, JavaScript
• **Scripting & Automation:**	Python (Basic workflow automation & data parsing).
• **Tools & DevOps:**	Docker, Git/GitHub, Linux/Ubuntu, VS Code, File I/O handling.
• **Databases & Storage:**	MySQL (Schema design, Query optimization), JSON local persistence, Redis (Basic Caching concepts).
• **Frameworks & Architecture:**	Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Layered Architecture, MVC, OOP/SOLID.
Soft Skills: • **Self-Leadership & Adaptability:**	Strong ability to self-manage, take ownership of tasks, and quickly adapt to new technology stacks (such as transitioning frameworks or languages).
• **Clear Communication & Presentation:**	Ability to articulate technical concepts clearly to team members and stakeholders.

- **Collaborative Teamwork & Problem-solving:** Experienced in technical troubleshooting and working effectively within cross-functional teams.

CERTIFICATIONS

MOS Certificate: Word, Excel, PowerPoint	2019
IELTS 6.0 (equivalent to TOEIC 650+).	2022
FUNiX Certificates: *Web/App Development* .	2025

REFERENCES

Lê Đức Thịnh (administrative officer)	Email: leducthinh@tdtu.edu.vn
Ms. Hannah Thảo Bùi - FUNiX,	039 550 6001
Nguyễn Thị Ngọc Dung (Deputy Directo)	0906365785

PROJECTS

AI Object Detection Integration | YOLO8, OPENCV 10-2025 - 2-2026

R&D

- Integrated YOLOv8 deep learning models with OpenCV pipelines to automate real-time video stream analytics.
- Designed structured data parsers to extract, format, and log localized object detection payloads for downstream systems.
- Optimized frame-processing workflows to ensure low-latency execution and stable runtime resource management.

Automotive Vehicle Profile & Telemetry Backend System | *Java, Spring Boot, Spring Data JPA, MySQL, Spring Security, JUnit 5* 10-2025 - 3-2026

Developer

- ***Description***:Built a robust RESTful API system designed to manage localized vehicle user profiles and telemetry settings (Display, Sound, General System states), simulating an enterprise-grade automotive data-management cluster
- ***Key Features***:
- **Robust REST API & Architecture:** Developed a clean, layered Spring Boot application separating Controller, Service, and Repository layers using SOLID principles.
- **Data Input & Validation:** Implemented Java Validation API (@Valid and Custom Validators) to filter and sanitize incoming vehicle metadata payloads (email formats, strict 8-digit unique IDs, numeric boundaries).
- **Persistence Layer:** Designed a relational MySQL schema mapped via **Hibernate/Spring Data JPA**, optimizing data query performance and configuring structural entity relationships.
- **Advanced Data Processing:** Leveraged **Java Stream API** to build high-performance dynamic sorting and filtering utilities, processing data clusters alphabetically or chronologically by timezone metrics.
- **Robust Exception Handling:** Implemented a @ControllerAdvice global exception handling mechanism to gracefully intercept runtime syntax errors or missing configurations, ensuring 99.9% system uptime stability.

INTERESTS

Gym, running, music, and traveling

ADDITIONAL INFORMATION

- **AI & Next-Gen Productivity:** Leverage Cursor AI, Claude, and local AI Agents (utilizing LLMs, Local AI, and MCP - Model Context Protocol) to automate **multi-threaded Java debugging, JVM thread dump analysis**, maintain README documentation, and enforce **Java coding standards (Google Java Style/Sonarlint)**, significantly accelerating software development workflows and increasing engineering efficiency.